HORIZON-GATED LATENT PREDICTION: DISENTANGLING PERSISTENT FACTORS IN TIME SERIES

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ABSTRACT

A long-lasting factor in a time series need not be a slow one: a factor can persist while riding on a nonlinear statistic such as an oscillation's frequency, where no low-pass reading reaches it. What persistent factors of either kind share is that they stay useful for predicting the distant future, while short-lived ones do not. We introduce Horizon-Gated Latent Prediction (HGLP), a self-supervised representation learning method for time series that disentangles long-lasting *persistent* factors from short-lived *transient* ones by gating the latent space as a function of the prediction horizon. A soft gate prevents long-horizon predictions from using a designated subspace (z_{mix}), forcing the model to route long-range predictive information through a specific block (z_{per}); a cross-covariance penalty and a per-block LayerNorm further encourage separation. We also propose a block-by-factor evaluation protocol and a single-number Separation Index (SEP), and demonstrate that the method separates factors on synthetic data and, more mixedly, on two real datasets (PTB-XL ECG, HAPT activities), with ablations showing the necessity of both the gate and a latent target. The penalty's one free coefficient can be set with no task labels at all.

1 INTRODUCTION

A sensor stream that runs unattended almost always carries two things at once. A wearable accelerometer records both what the wearer is doing and how the device happens to be shaking in this instant. The quantity anyone acts on is the first — the state that lasts — and the second changes far faster than any decision taken from it. This is the ordinary situation in condition monitoring, activity recognition, sleep staging and health-state estimation, and it is why the two are worth keeping in separate coordinates. A named block buys three things: interpretation gets a fixed address rather than a post-hoc story, the price of using only that block becomes a measurable number, and scarce labels can be spent on a narrow subspace instead of the whole embedding.

Why the separation has to happen during training. An embedding can always be rotated afterwards, and unsupervised unmixings — principal component analysis, slow feature analysis (SFA), independent component analysis — do exactly that at a fraction of the cost, so this is the sharpest objection to anything done at training time. A post-hoc method must rank directions by a quantity computable from a frozen embedding: variance, slowness, non-Gaussianity, or predictability under a linear autoregressive model (Richthofer & Wiskott, 2013). **None of those is persistence.** The horizon is not a property of the embedding at all but of the predictive objective: free while training, and gone once training ends. Section 5.4 measures what that buys and what it does not — where the surrogate and persistence agree, which is the ordinary case on real data, post-hoc rotation is as good or better.

Persistence is not confined to the slow band. Separating "slow" from "fast" can mean separating the *signal*, decomposing a waveform into trend and fluctuation (Cleveland et al., 1990; Huang et al., 1998), or separating the *factors* that generated it (Hyvärinen & Morioka, 2016) — a patient's diagnosis, or the fact that a person is walking. The two readings are routinely treated as one, and they diverge in exactly one direction. Fixing the term: a factor is *persistent* if it still contributes to prediction Δ steps ahead. The slow part of the signal is then necessarily persistent, since a component confined to low frequencies has slowly decaying autocorrelation (Khinchin, 1934) and remaining autocorrelation is a contribution to distant prediction. The converse fails, because low-pass filtering is linear in the signal while a factor may ride on a nonlinear statistic of it. Engineering practice knows this: in rolling-element bearing diagnostics the fault frequencies do not stand out in the spectrum of the raw vibration and become apparent only after demodulation, in the spectrum of the Hilbert envelope (Randall & Antoni, 2011); in frequency-modulated radio the message lives in the instantaneous frequency of a fast carrier, where no low-pass of the raw waveform reaches it. Our synthetic generator is built on that second shape deliberately, and Section 3 shows what it costs a slowness criterion. This is an existence claim, not a claim about how often such factors arise.

One criterion covers both kinds. So a method keyed to the slow band reaches persistent-and-slow factors and misses persistent-but-not-slow ones. What both kinds share is the property they were defined by: **both remain useful for predicting the distant future, and transient factors do not.** That criterion is controlled by a variable prediction-based learning already contains — the horizon Δ, since pushing it out leaves only long-persisting factors useful (Bialek et al., 2001; Creutzig & Sprekeler, 2008) — and nothing in that argument refers to which band the factor occupies. We gate that axis. HGLP places a single soft threshold on the horizon: beyond it the predictor may no longer read a designated block, so the only remaining route to lowering long-range loss is to write the current state into the block that is left. Because the latent space is rotation-symmetric, the persistent factor can be read from any 16 dimensions, and our claim is therefore not that it *gathers* in z_{per} but that the transient factor is *emptied out* of it; the complement z_{mix} is left unconstrained and keeps both (Section 4.2).

Contributions. (i) An inductive bias that gates an axis already present in any predictive learner, splitting the persistent factor into a designatable block, together with a criterion that sets the one coefficient governing that split *without task labels*. (ii) An evaluation protocol that scores the separation from both sides — a block $\times$ factor matrix — and a single summarizing index, SEP. (iii) Measurements on synthetic data, PTB-XL and HAPT, including an audit of where the method fails. The results cut both ways. On synthetic data we lead label-free post-hoc rotation on both loss families (SEP 0.849 against SFA's 0.388); on real data we trail SFA in three of four settings by 0.031–0.072. Splitting the index into its three terms locates that shortfall entirely on *exclusion*, the one term our objective cannot enforce, and replacing the linear probe with a small neural one — given equally to every baseline — collapses that same term and no other. The separation also costs -0.111 macro-F1 downstream on HAPT.

2 Related Work

2.1 Classical Methods

Classical methods apply a *fixed transform* to the raw signal to read out its factors, and the dominant line has been the search for slowly varying components, since the slowly varying part of a signal reflects the factors that persist. The most direct approach splits the waveform itself: STL decomposes a series into seasonal and trend components (Cleveland et al., 1990), and EMD extracts oscillatory modes empirically (Huang et al., 1998). These deliver *components of the signal*, not the factors that generated it.

Another strand optimizes the slowness of the output directly. Slow Feature Analysis (SFA) minimizes the temporal derivative of the output under unit variance and pairwise decorrelation constraints, and, since the input is expanded into low-order polynomials, it also extracts nonlinear features (Wiskott & Sejnowski, 2002). Strengthening decorrelation into statistical independence makes the slow components separate from one another too, combining slowness with independent component analysis (Blaschke et al., 2007). We use these two variants as our label-free post-hoc rotation baselines. A further strand changes the criterion: Predictable Feature Analysis (PFA) keeps SFA's extraction scheme but selects the most predictable features instead of the slowest, predictability being how well the next value is approximated from the most recent p values (Richthofer & Wiskott, 2013). Yet another changes where the signal is read: the Fourier transform reaches factors riding on the *modulation* rather than the values, as in the envelope analysis of rotating-machinery diagnosis (Randall & Antoni, 2011).

What this family has in common is that *the transform is not learned*: which statistics and bands to keep are fixed in advance by a human, and the projection is refitted over the whole signal whenever the domain changes.

2.2 Self-Supervised Learning for Time Series

Time-series representation learning trains an encoder mapping a signal segment to a vector without labels, taking its learning signal from the temporal structure of the data—neighboring segments should be similar, or what is observed now should predict what comes shortly after. The family splits into three groups by the *form of the target*.

The *reconstruction family* is the largest and effectively the default. Autoencoders pass a signal through a narrow bottleneck and restore it, the remaining error itself being useful for anomaly detection, imputation, and forecasting pretraining. Variational autoencoders add a prior over the latent variables, and that prior becomes a *handle for inducing structure in the latent space*: a representative variant weights the KL term to push the coordinates toward independence (Higgins et al., 2017), a lineage leading to Section 2.3. Masked pretraining, which reconstructs randomly masked patches, has since become established (Nie et al., 2023).

The *contrastive family* decides which pairs count as similar and pulls and pushes the representations accordingly. CPC predicts the future contrastively in the latent space (van den Oord et al., 2018), and later time-series methods refined those pairs: contrasting hierarchically over augmented context views (Yue et al., 2022), or using local smoothness to define stationary temporal neighborhoods (Tonekaboni et al., 2021). Their weakness is that what gets discarded is hidden in the pair-selection rule.

The *latent-prediction family* takes the representation itself, not the raw signal, as the target (LeCun, 2022; Assran et al., 2023). Its time-series variants differ in where the target is read: from masked patches (Ennadir et al., 2025), from the next latent token (Chemeris et al., 2026), or from an attention-pooled summary of the whole future interval (Petersen et al., 2026).

This work belongs to the latent-prediction family but uses *both* a regression-form (L_1) and a contrastive-form (InfoNCE) loss, reporting the two stems together to show that the horizon gate does not rely on a particular loss form. All three groups *learn* the representation, but the objective does not determine what each coordinate means.

2.3 Disentangled Representations for Time Series

Work here has been most successful at separating *factors constant over an entire sequence* from the rest, because invariance transfers directly into an objective. FHVAE separates what stays fixed within a sequence from what changes across segments by assigning different priors (Hsu et al., 2017), and the disentangled sequential autoencoder splits the latent into static and dynamic parts (Li & Mandt, 2018). On identifiability, exploiting non-stationarity by learning to distinguish time segments makes a nonlinear ICA model identifiable *up to a pointwise transformation of each factor* when combined with linear ICA (Hyvärinen & Morioka, 2016), and an approach placing a *sparse* prior on the change between adjacent observations followed (Klindt et al., 2021). That models cannot be selected without labels (Locatello et al., 2019) applies equally to our hyperparameters, and we address it in the limitations.

What this family leaves empty is *factors that vary slowly but are not constant*: the static/non-static dichotomy pushes them into the "dynamic" side and separates them no further. None of the three strands creates coordinate-block structure of the kind we seek; this work fills that place with the axis of the prediction horizon.

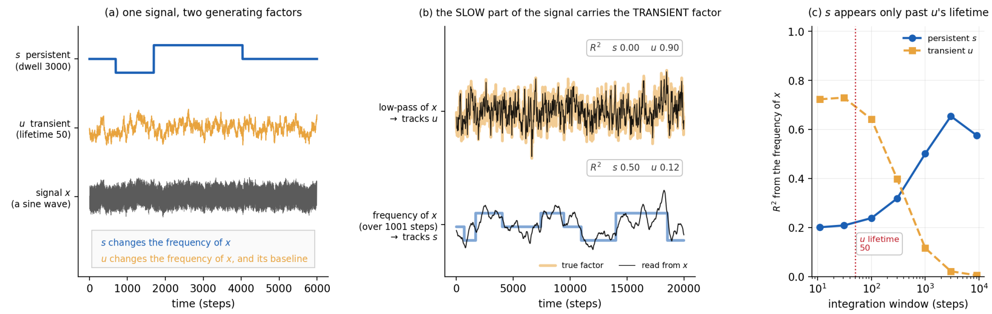

Figure 1: **Spectral slowness is not factor persistence.** One synthetic series, 400,000 steps, one seed. **(a)** The two factors and the signal they generate. **(b)** The slow part of x against the frequency of x read over a 1,001-step window, each scored on both factors. **(c)** The same R^2 against the width of that window.

3 Background: where persistence is not slowness

If persistence and spectral slowness always agreed, a horizon gate would be an expensive way to reach what a low-pass filter already gives. We therefore build data in which the two disagree by construction.

A three-state Markov persistent factor s (mean dwell 3,000 steps) sets the carrier frequency of an oscillation, spaced $\pm 3\%$ around 0.100; a transient Ornstein–Uhlenbeck factor u (autocorrelation time 50 steps) modulates that instantaneous frequency and adds to the signal. Amplitude is identical across regimes, so a regime change alters *only how fast the signal oscillates*. The two timescales are cleanly separated — autocorrelation half-lives 1,859 and 51 steps — so s is persistent by any reading. Yet it is absent from the slow part of the signal: a low-pass reading ($f < 0.02$) recovers the *transient* factor at $R^2 = 0.90$ and the persistent one at $R^2 < 10^{-5}$ (Fig. 1b). The persistent factor is carried by the fastest-changing part of the signal and the transient factor enters as the slower additive term, which is the exact reversal of what a slowness criterion assumes.

The persistent factor is still readable, but only through a nonlinear statistic and only over a wide enough window: the instantaneous frequency over 1,001 steps recovers it at $R^2 = 0.50$, and the recovery appears only once the window is tens of times u's lifetime, falling away again when the window grows wide enough to blur s itself (Fig. 1c). This is the general shape of the situation, not an artifact of our generator: while a factor persists, what stays fixed is not the signal but its statistics, and those surface only through some function of the signal.

Difficulty is set by the spacing between regime frequencies and calibrated by a training-free FFT reader, so that the task is neither trivially solved nor unsolvable before any representation is learned. At $\pm 3\%$ the reader reaches accuracy 0.660 (chance 0.333); at $\pm 5\%$ it already reaches 0.784 before any training, and at $\pm 1\%$ it falls to 0.459, near chance. We use $\pm 3\%$ throughout.

4 Method

4.1 Model

Encoder. A patch becomes a token through one linear projection, so channels are mixed at the first layer — HAPT posture is defined by how gravity distributes over the three axes. Tokens pass through causal transformer layers with rotary position encoding, and a final linear layer produces the embedding, whose leading 16 coordinates form z_{per} and the remaining 48 z_{mix}. A per-block LayerNorm is applied last, to the two blocks separately rather than to the embedding as a whole (Section 4.3).

Predictor. A small MLP reading the embedding at one time point together with the horizon, which enters as a learned expansion of $\log_2 \Delta$. One predictor is conditioned on Δ rather than one head per horizon.

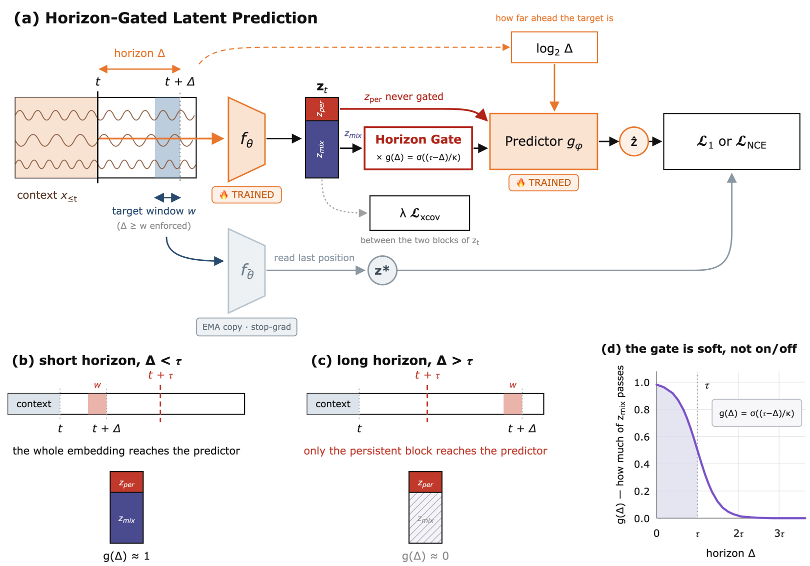

Figure 2: Horizon-gated latent prediction. **(a)** A causal encoder maps a window to z_t, split into z_{per} (16 of 64 dimensions) and z_{mix}; a predictor conditioned on Δ matches a target read at one position inside $(t,\, t+\Delta]$. **(b, c)** The same pipeline at a short and a long horizon. **(d)** The gate, with the axis in units of τ.

The attention operation is unmodified: the gate sits between encoder output and predictor input, the penalty in the loss, the per-block LayerNorm in the encoder's last line. This is a change of training procedure, not of architecture, so it attaches to any causal sequence encoder.

Target.

$$z^* = \mathrm{sg}\big[\, f_{\bar\theta}\big(x_{[t+\Delta-w,\; t+\Delta]}\big)_{-1}\big], \qquad \bar\theta \leftarrow m\bar\theta + (1-m)\theta \tag{1}$$

The prediction target is not the raw signal but the representation an EMA copy of the encoder reads from a short window inside the future interval, with the gradient stopped. Bounding that window keeps the target's receptive field inside $(t,\, t+\Delta]$, so it never re-contains the anchor's own past and the shortcut of matching the target from what is already known at the anchor is closed by construction. **The target never passes the gate and is always the full embedding**: at far horizons the model is asked to match the whole future embedding from the persistent block alone, and this asymmetry is the core of the mechanism.

Loss.

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda\, \mathcal{L}_{\text{xcov}}, \qquad \mathcal{L}_{\text{pred}} \in \{\mathcal{L}_1,\ \mathcal{L}_{\text{NCE}}\} \tag{2}$$

The two forms share backbone, gate and penalty and differ only in the loss layer. **HGLP-Reg** uses the L_1 distance; under L_2 the squared-error gradient is dominated by large residuals and results swung across seeds. **HGLP-NCE** uses InfoNCE over cosine similarity with temperature 0.1, with one departure from the standard form: **samples from the same window are excluded from the negative pairs**, since they share the persistent factor and pushing them apart would teach the model to discard it.

4.2 The Horizon Gate

$$g(\Delta) = \sigma\Big(\frac{\tau-\Delta}{\kappa}\Big), \qquad \tilde z_t(\Delta) = \big[\, z_t^{\text{per}}\,;\; g(\Delta)\, z_t^{\text{mix}}\,\big] \tag{3}$$

Immediately before the predictor, the rear block is multiplied by $g(\Delta) \in (0,1)$ (Fig. 2(b)–(d)): for $\Delta \ll \tau$ the embedding passes unchanged, for $\Delta \gg \tau$ the rear block is suppressed. *Dimensions are multiplied, not truncated*, so the predictor input is always D-dimensional; z_{per} *is never* *blocked*; and the gate is *a smooth curve, not a step*. The trained horizons must straddle the threshold, $\Delta_{\min} < \tau < \Delta_{\max}$, or the gate never opens or never closes. The width κ is a buffer rather than a convenience, since τ cannot be chosen accurately without labels, and a step gate was worse on both stems (Appendix A).

The one-sidedness is likewise a choice. Withholding z_{mix} beyond τ should cost nothing, because the transient factor is already forgotten there. Withholding z_{per} below τ would cost a great deal: near horizons need the current persistent state as much as the fast detail, and they are where the gradients are dense, so the block would lose most of its training signal rather than gain protection. Appendix A tests the symmetric variant. Note also that only the *availability* of z_{mix} is constrained, never its content — across the six settings it carries the persistent factor at 0.707–0.879 — which is why we call it z_{mix} and not z_{tra}.

Gradient split.

$$\frac{\partial \mathcal{L}}{\partial z_t^{\mathrm{per}}} = \frac{\partial \mathcal{L}}{\partial \tilde{z}_t^{\mathrm{per}}}, \qquad \frac{\partial \mathcal{L}}{\partial z_t^{\mathrm{mix}}} = g(\Delta)\,\frac{\partial \mathcal{L}}{\partial \tilde{z}_t^{\mathrm{mix}}} \tag{4}$$

Because the encoder is shared across horizons, this one line splits the training signal the two blocks receive: z_{per} is updated at every horizon, whereas z_{mix} is updated in proportion to $g(\Delta)$ and so effectively only at near horizons. Two consequences follow, and only one of them is guaranteed. **Availability is enforced**: whatever still contributes to prediction beyond τ must be readable in z_{per}, since no predictor however large can restore information the gate has suppressed to zero. **Exclusion is not enforced**: z_{per} is still used at near horizons, where transient information lowers the loss, and no term of the objective pushes that information *out of* z_{per} — exclusion has to come from the penalty of Section 4.3. We therefore measure exclusion rather than assume it, and Section 5.4 follows that measurement through.

4.3 Pulling the Two Blocks Apart

Cross-covariance penalty.

$$\begin{aligned} C &= \tfrac{1}{N-1}\,\bar{Z}_{(1:d_{\mathrm{per}})}^{\top}\,\bar{Z}_{(d_{\mathrm{per}}+1:D)}, \\ \mathcal{L}_{\mathrm{xcov}} &= \tfrac{1}{d_{\mathrm{per}}(D-d_{\mathrm{per}})}\textstyle\sum_{a,b} C_{ab}^2 \end{aligned} \tag{5}$$

The penalty pushes the two blocks to be linearly unpredictable from each other and **acts on the embedding before the gate**. It travels with the **per-block LayerNorm**: a single LayerNorm over the whole embedding gives the two blocks one shared mean and variance, recreating exactly the coupling the penalty is trying to remove. **"Mechanism-free" therefore means all three devices absent** — gate, penalty and per-block LayerNorm.

4.4 What We Measure

Reading the persistent factor from z_{per} is not by itself evidence of separation, since rotational symmetry means it is read about as well from any 16 dimensions. We therefore report a **block $\times$ factor matrix** scoring both blocks on both factors, plus a single-number summary **SEP**, which asks three questions.

1. **Inclusion — is the persistent factor in z_{per}?** We score how well the persistent factor is predicted from z_{per} alone.
2. **Allocation — is the transient factor in z_{mix}?** We score how well the transient proxy is recovered from z_{mix} alone.
3. **Exclusion — is the transient factor absent from z_{per}?** We measure how well the transient proxy can be recovered from z_{per}, and subtract that value from 1.

$$\mathrm{SEP} = \underbrace{\big[s_{\mathrm{per}}(z^{\mathrm{per}})\big]_0^1}_{\text{inclusion}} \cdot \underbrace{\big[s_{\mathrm{tra}}(z^{\mathrm{mix}})\big]_0^1}_{\text{allocation}} \cdot \underbrace{\big[1 - s_{\mathrm{tra}}(z^{\mathrm{per}})\big]_0^1}_{\text{exclusion}} \tag{6}$$

Table 1: The three datasets. The transient proxy is computed from the signal, never from labels. Channels are 1, 1 (lead II) and 3 (acceleration axes); the grouping unit for splits is 4,812 patients for PTB-XL and 30 subjects for HAPT.

	Size	Rate	Patch	Window L	Persistent factor	Transient proxy
Synthetic	10^6 steps	—	8 steps	256	regime s, 3-way	OU u
PTB-XL	5,000 rec.	100 Hz	10 (0.1 s)	100 (10 s)	diagnosis, 2-way	voltage
HAPT	2,104 win.	50 Hz	4 (0.08 s)	256 (20.5 s)	activity, 6-way	$\|a\|$

Table 2: Separation index by mechanism. Cells are SEP, $n = 5$ seeds, $\lambda = 4$; bold marks the best cell in each row.

Dataset	Stem	Neither	Gate only	xcov only	**Gate + xcov**
Synthetic	Reg	0.135	0.342	0.279	**0.408**
Synthetic	NCE	0.121	0.124	0.488	**0.749**
PTB-XL	Reg	0.375	0.404	0.312	**0.495**
PTB-XL	NCE	0.392	0.503	0.283	**0.567**
HAPT	Reg	0.521	0.537	0.236	**0.632**
HAPT	NCE	0.297	0.366	0.162	**0.515**

$s_{\text{per}}(\cdot)$ scores reading the *persistent factor* from a subspace and $s_{\text{tra}}(\cdot)$ the *transient proxy*; $[\cdot]_0^1$ clips into $[0, 1]$. Classification scores already lie in this range, but R^2 goes negative when a prediction is worse than the mean, and the clipping acts only there, preventing a negative value from flipping the sign of the product. There is no normalization by any denominator anywhere.

Probe protocol. With the encoder frozen we fit only a linear classifier or regressor. Splits are **per group** (subject for HAPT, patient for PTB-XL, a front/back time split for the synthetic data), evaluation windows are **non-overlapping** to prevent leakage (asserted in the code), and only positions after the **minimum context length (64 patches)** are scored, since with less context the encoder lacks the evidence to read the persistent state. A **random subspace** of the same width is scored as a control.

5 Experiments

Every setting below is one of six: three datasets crossed with the two loss stems, at $n = 5$ seeds unless stated. We ask four questions in order — what produces the separation, whether its one free coefficient can be set without labels, how it compares against label-free post-hoc rotation, and what it costs.

5.1 Data and factors

We use the synthetic generator of Section 3 and two public sets, PTB-XL (Wagner et al., 2020) and HAPT (Reyes-Ortiz et al., 2016; 2015); Table 1 gives their sizes and their two factors.

Splits follow the protocol of Section 4.1, with the patient as unit for PTB-XL and the subject for HAPT. No data is selected by label before training: HAPT's transition and unlabeled segments, 24.7% of the total, are excluded from scoring but left in the pre-training input, windows are cut without consulting activity boundaries, and labels are read per position. On PTB-XL the rule that initializes τ does not apply, since the heartbeat repeats regularly and its autocorrelation oscillates instead of decaying; τ is set by hand there (Section 6.1).

5.2 What creates the separation

We switch each of the two devices we add — the horizon gate and the cross-covariance penalty — on and off. The mechanism-free cell (Section 4.3) is the baseline in which only the coordinate split remains.

In Table 2 the model with both devices ranks first in all six settings, and neither device alone reaches that value. On the contrastive stem of the synthetic data the gate alone is indistinguishable from the

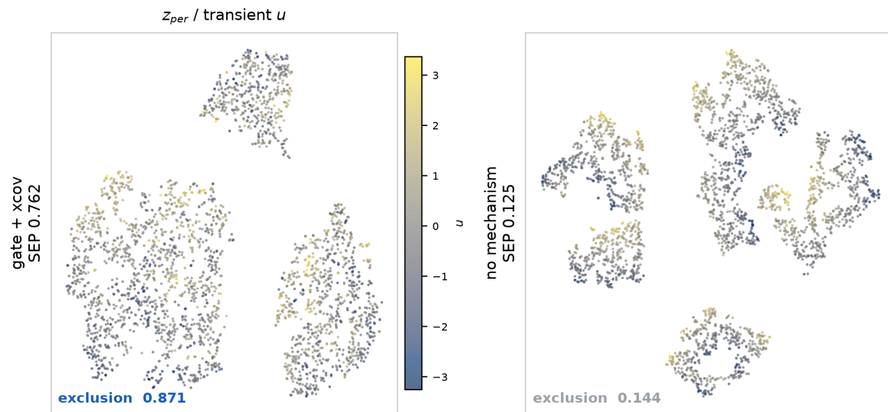

Figure 3: The exclusion term, seen directly: a t-SNE of z_{per} coloured by the transient factor, with the mechanism and without it (synthetic $\pm 3\%$, HGLP-NCE, $\lambda = 4$, one seed). This is the only one of the three terms that differs between the two models — inclusion is 0.981 against 0.983 and allocation 0.891 against 0.880. All six panels are in Fig. 6.

mechanism-free case (0.124 against 0.121), yet adding that same gate to the penalty raises SEP from 0.488 to 0.749: the two devices need each other. The penalty alone is actively harmful on the real data, breaking down the inclusion term and falling below even the mechanism-free case.

The "Neither" cells take the first 16 of the 64 dimensions, and their SEP agrees in all six settings to within 0.015 with the SEP of 16 dimensions drawn at random from the same embedding. Without our devices, any part of the embedding holds the two factors to a similar degree, so coordinate position carries no meaning by itself; our contribution is to give those coordinates structure. Fig. 3 shows the same contrast qualitatively.

Three further design choices could each have gone the other way, and none of them is left to argument. Replacing the smooth gate with a step at τ, gating z_{per} symmetrically as well, and predicting the raw signal instead of a latent target are each tested in Appendix A; all three are worse, and the latent target by a factor of two.

5.3 Setting the penalty weight without labels

λ was chosen by looking at downstream performance, which makes "our best λ separates well" partly circular. We therefore define a criterion that uses no task label at all, the **purity–persistence score**: z_{per} should be *pure*, holding none of the transient factor, and *persistent*, still predicting its own value far ahead.

$$\text{PPS} = \underbrace{\big[1 - s_{\text{tra}}(z^{\text{per}})\big]_0^1}_{\text{purity}} \cdot \underbrace{s_{\text{self}}(z^{\text{per}};\, \Delta > \tau)}_{\text{persistence}}, \tag{7}$$

Purity is one minus the transient proxy's score from z_{per} — the exclusion term of Eq. 6, and the proxy is computed from the signal. Persistence is how well z_{per} predicts *its own* value more than τ ahead; neither the labels nor the target encoder enters. The product is required because each half alone has a degenerate maximum: an empty block is perfectly pure, a constant block is perfectly self-predicting, and multiplying kills both.

Selecting λ by PPS reproduces the label-selected value in all six settings (Table 3). On synthetic data the criterion is also doing real work rather than agreeing by luck: purity alone would have chosen $\lambda = 16$, close to the worst choice by the oracle (SEP 0.332 against 0.814 at $\lambda = 1$), and it is the persistence half, collapsing from 0.670 to 0.051 once $\lambda \geq 4$, that rules it out. A collapse monitor would not have caught this: effective rank *rises* from 4.3 to 12.3 over the same range, so the penalty is not emptying z_{per} but scattering it along the time axis.

The margin between the top two PPS values is 0.111–0.234 on synthetic but only 0.004–0.023 on the real sets, and we did not store its seed-level spread, so the four real-data agreements are not

Table 3: Choosing λ without labels: the two columns agree in every setting. The oracle is the λ maximizing SEP, which does use labels. $n = 3$ seeds; the margin is the gap between the top two PPS values, and a small margin is not confident agreement.

Dataset	Stem	λ by PPS (no labels)	λ by SEP (oracle)	PPS margin
Synthetic	HGLP-Reg	1	1	0.234
Synthetic	HGLP-NCE	16	16	0.111
PTB-XL	HGLP-Reg	4	4	0.023
PTB-XL	HGLP-NCE	4	4	0.004
HAPT	HGLP-Reg	1	1	0.016
HAPT	HGLP-NCE	64	64	0.009

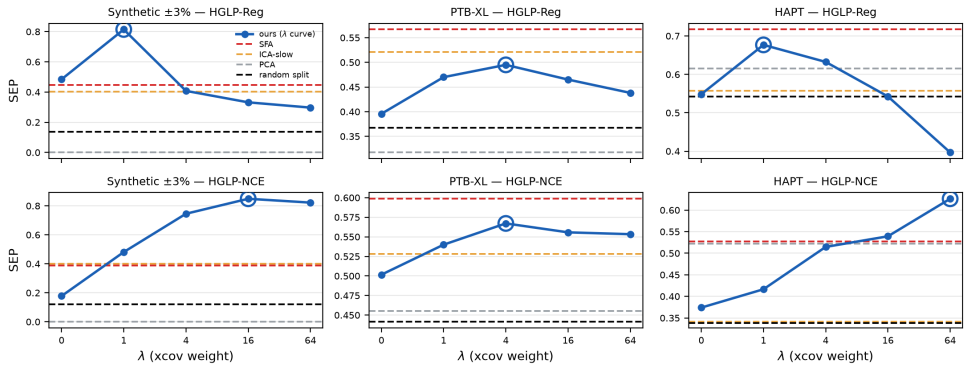

Figure 4: SEP against the penalty weight λ, $n = 5$ seeds. Our method traces a curve with its best point ringed; each post-hoc unmixing has no knob and is a horizontal line. A win is our curve rising above the SFA line.

distinguishable from chance at this sample size. The honest reading is that the criterion is verified on synthetic data, consistent on real data, and cheap to run wherever the transient proxy exists.

5.4 Against label-free post-hoc rotation

If an embedding trained without the gate is rotated post-hoc without labels — by principal component analysis, by slowness-ordered independent component analysis, or by slow feature analysis (SFA) — does it reach the same separation? These methods need no horizon gate at all, so they are the sharpest objection to our method.

We have one knob and the post-hoc rotations have none, so we report the whole curve rather than a single point (Fig. 4). On the synthetic data the curve passes above every post-hoc rotation point, and principal component analysis fails completely (SEP 0.000) — direct evidence that a ranking by variance is not a ranking by persistence, since the transient factor crowds into the leading components and nothing is left in the complement. On the real data we do not lead. The verdict over all six settings is three wins to three losses, but split apart it is two of two on synthetic and three losses of four on real data, with losing margins of 0.031–0.072 against a winning margin of 0.099.

Where the shortfall sits. SEP is a product of three factors, so a single number cannot say which one is losing. Splitting it and pairing by seed gives an unusually clean answer (Appendix B, Table 6): inclusion is a draw in all six settings (-0.041 to $+0.102$), **every real-data loss is a loss on exclusion alone**, and every win is a win on allocation. The method does not degrade diffusely; it degrades on the one term the objective was never able to enforce (Section 4.2).

The exclusion we do achieve is a linear-level one. Every probe so far has been linear, and an unreadable direction is not the same as an absent one. Holding the features, the splits and the blocks fixed and replacing only the reading head with a one-hidden-layer MLP — given equally to SFA, PCA and ICA — exactly one term collapses, and it is exclusion: averaged over four methods and four real settings, inclusion moves $+0.015$ and allocation $+0.087$, while exclusion falls -0.149 and

Table 4: Degradation under domain shift. Each cell is the gated block's drop in macro-F1 from the clean domain to the most perturbed one, minus the same drop for a width-matched mechanism-free block; paired by seed, $n = 5$. Negative means it degrades less — **it never overtakes in absolute terms**.

Setting	25 labels	100 labels	All labels
HAPT / Reg	$\mathbf{-0.113{\pm}0.023}$	$\mathbf{-0.096{\pm}0.029}$	$-0.053{\pm}0.056$
HAPT / NCE	$-0.017{\pm}0.072$	$+0.015{\pm}0.012$	$+0.007{\pm}0.016$
PTB-XL / Reg	$-0.044{\pm}0.048$	$-0.033{\pm}0.048$	$-0.024{\pm}0.035$
PTB-XL / NCE	$-0.036{\pm}0.032$	$-0.010{\pm}0.024$	$-0.009{\pm}0.024$
Synthetic / Reg	$-0.012{\pm}0.102$	$+0.043{\pm}0.099$	$+0.064{\pm}0.077$
Synthetic / NCE	$-0.044{\pm}0.069$	$-0.055{\pm}0.054$	$-0.045{\pm}0.055$

as much as -0.530 in the worst row. The transient information is still present in z_{per}; the linear probe could not reach it. Our claim therefore narrows to a linear-level one — and the diagnosis above is not an artifact of weak probing, since SFA's lead on exclusion survives the stronger head and in places widens, while PCA and ICA degrade sharply. Appendix B gives the per-setting values and the caveats.

Why SFA leads where it leads. The explanation is that persistence and slowness mostly coincide on real data. On PTB-XL this is plain: the diagnosis is constant within a record, so its time derivative is zero, which is the limiting solution of the quantity SFA maximizes. On HAPT it is not constant within a window and we did not measure why the coincidence holds there, so that half of the explanation remains untested. What the synthetic data shows is the other side of the same statement: built so that the persistent factor is neither constant within a window nor present in the slow band, it is the one dataset on which the post-hoc route fails. What remains ours is that the coordinates are fixed during training, so no second fitting stage is needed at deployment — an operational difference, not a difference in the quality of the separation.

5.5 The cost of separation, and stability under domain shift

Measured against the full embedding of a mechanism-free model — not against our own z_{full}, which would mix in the effect of block width — the downstream cost of using z_{per} alone is effectively zero on synthetic data and PTB-XL, and -0.111 macro-F1 on HAPT (regression stem; -0.072 contrastive). HAPT is the only dataset whose persistent factor is not constant within the window: 57% of its windows straddle an activity transition, which is also why its cost is the one that shows.

We then carry a probe fitted on clean data with few labels into a domain where the transient factor has been perturbed by a gain, $x \mapsto x \cdot (1 + s)$ up to $s = 2$ — a bijection that leaves the persistent labels legible (HAPT activities survive at macro-F1 0.696, chance 0.167). The effect is small and mostly inside the noise. At 25 labels on HAPT/Reg the gated block's drop is 0.113 smaller than a width-matched mechanism-free block's, five times the seed standard deviation; in the other five settings the effect lies inside the noise, and 17 of 24 cells are negative by sign alone (Table 4). Nowhere does absolute performance overtake the mechanism-free model. HAPT is at once the dataset with the largest cost and the only one with a visible shift effect: the gate buys a smaller drop and pays for it in absolute performance.

6 Conclusions

The horizon gate and the cross-covariance penalty split the persistent factor into a designated coordinate block, and across all six settings formed by three datasets and two loss families they jointly produce a separation that neither device reaches alone. The separation requires a latent target: predicting the raw signal instead halves the score. On synthetic data, built so that persistence and slowness disagree, we exceed label-free post-hoc rotation for both loss families; on real data, where the two mostly coincide, we fall short of Slow Feature Analysis in three of four settings, and the shortfall sits entirely on the term the objective cannot enforce. The coefficient governing that term can be set without task labels.

6.1 LIMITATIONS

The exclusion we demonstrate holds at the linear level only. The transient factor is not absent from z_{per}; it is linearly unreadable there (Section 5.4). A structural repair does not help either — a gate that also switches z_{per} off below τ is worse in 4 of six settings.

Most hyperparameters cannot be chosen without labels. The block width, the target window, the patch size and the range of training horizons have no derivation rule and were set by downstream performance, a known difficulty in this literature (Locatello et al., 2019); the target window in particular governs how much of the transient factor is averaged away inside the target, and we never swept it. The threshold τ can at least be initialized from the transient factor's measured autocorrelation time, but that rule fails on regularly repeating signals — on PTB-XL the per-record estimates spread over a factor of 499 — so it is set by hand there. The cross-covariance weight is the one knob that is genuinely settled: the purity–persistence score (Eq. 7) agrees with the label-based choice in all six settings, though decisively only on synthetic data, where the margin to the runner-up is 0.111–0.234 against 0.004–0.023 on real data.

We did not demonstrate a practical benefit. The perturbed-domain effect clears the error bars in one of six settings, and absolute performance never overtakes the mechanism-free embedding.

Our axis selects persistence, not relevance, and the case for it rests on data we built. A persistent nuisance factor — the low-frequency drift respiration and electrode motion produce in an electrocardiogram — enters z_{per} by the same criterion, and which persistent factors matter remains a human judgment. The claim that a factor can be persistent yet absent from the slow part of the signal likewise rests on one synthetic dataset: a single counterexample does refute a universal proposition, but we designed this one ourselves.

6.2 FUTURE WORK

We compared only against label-free post-hoc rotation, so methods that place factor separation explicitly in the objective remain to be compared — as does predictable feature analysis (Richthofer & Wiskott, 2013), the post-hoc criterion closest to persistence and the one our argument most owes a measurement. Validation is also needed on data whose persistent factor varies within a recording while staying constant within a window: PTB-XL fails the first property and HAPT the second, whereas sleep-stage classification has both. We have that fourth dataset in hand and the analysis is under way; it is unreported because its window-length axis has not been tested, and on it the label-free criterion selected a different coefficient from the label-based one. Finally, reading the trajectory of the separated block as a state indicator rather than as classifier input (Appendix C) would address the third limitation directly.

7 WHAT IS NEW IN THIS GENE 501 REPORT

An earlier version of this work, written for another course, proposed the horizon gate, the cross-covariance penalty and the SEP protocol, and reported the mechanism ablation, the post-hoc rotation comparison, and the downstream cost and shift results. This report adds:

- **Label-free selection of the penalty weight** (Section 5.3). The earlier version set λ by downstream performance. The purity–persistence score uses no task label and recovers the same value in all six settings. This is the only change to the method itself.
- **Ablations of three design choices** (Appendix A): the smooth gate against a step gate, the one-sided gate against a symmetric one, and the receptive-field-bounded target against the cumulative one. The earlier version defended all three in prose.
- **A non-linear probe** (Appendix B): the same MLP reading head given to our method and to every baseline, which shows that the exclusion the method achieves holds at the linear level only.

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A Design choices, tested

Everything here holds the rest of the model fixed and changes one choice.

The gate is smooth. A step gate, $\mathbf{1}[\Delta < \tau]$, zeroes $z_{\rm mix}$'s gradient outright beyond the threshold rather than passing a partial one. On synthetic data it lowers SEP from 0.408 to 0.353 on HGLP-Reg and from 0.749 to 0.684 on HGLP-NCE ($n = 5$ seeds). Since τ cannot be chosen accurately without labels, horizons near it should not flip between fully passed and fully blocked under the smallest movement of the threshold, and the width κ is what gives them intermediate values.

The gate is one-sided. The symmetric variant multiplies $z_{\rm per}$ by $1 - g(\Delta)$, switching it off below τ so that no horizon can deposit transient information in it — the obvious structural way to buy the exclusion the objective cannot enforce. It does not work (Table 5): better in 1 of six settings and worse in 4, costing 0.141 SEP on the synthetic regression stem. Removing $z_{\rm per}$ from the near horizons removes its training signal there as well.

Table 5: Can exclusion be bought structurally? The symmetric variant multiplies $z_{\rm per}$ by $1 - g(\Delta)$, switching it off below τ so that no horizon can push transient information into it. Cells are SEP at each model's own best λ over $\{0, 1, 4, 16, 64\}$, $n = 5$ seeds; higher is better; differences within ± 0.005 SEP are called a tie. The asymmetric gate of Eq. 3 is better in 4 of 6 settings against 1 for the symmetric one, so the design choice is not an oversight.

Dataset	Stem	Asymmetric (ours)	Symmetric	Difference	Better
Synthetic	HGLP-Reg	0.814 ($\lambda = 1$)	0.673 ($\lambda = 1$)	-0.141	asym
Synthetic	HGLP-NCE	0.849 ($\lambda = 16$)	0.873 ($\lambda = 4$)	$+0.024$	**sym**
PTB-XL	HGLP-Reg	0.495 ($\lambda = 4$)	0.438 ($\lambda = 1$)	-0.057	asym
PTB-XL	HGLP-NCE	0.567 ($\lambda = 4$)	0.567 ($\lambda = 16$)	-0.000	tie
HAPT	HGLP-Reg	0.677 ($\lambda = 1$)	0.575 ($\lambda = 4$)	-0.102	asym
HAPT	HGLP-NCE	0.626 ($\lambda = 64$)	0.598 ($\lambda = 64$)	-0.028	asym

The target's receptive field is bounded. The gate rests on a premise: withholding $z_{\rm mix}$ at long range should cost nothing, because the transient factor is already forgotten there. That premise is a property of the data *and of how the target is defined*, so it can be measured. For each horizon we fit two linear readings of the target embedding, one from the persistent factor alone and one from both, and take the difference (Fig. 5).

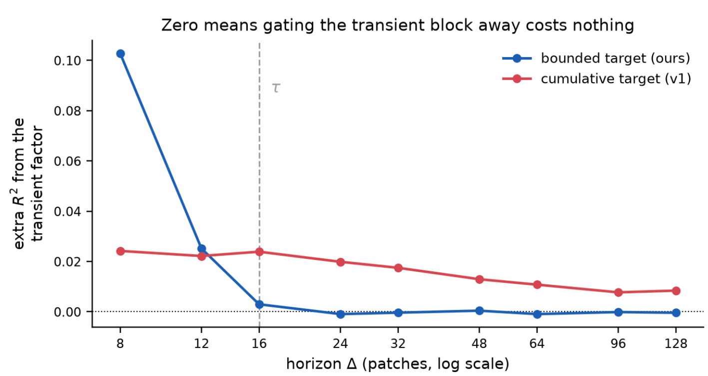

Figure 5: $R^2(s, u \to \bar{z}_{t+\Delta}) - R^2(s \to \bar{z}_{t+\Delta})$: what the transient factor adds, beyond the persistent one, to a linear reading of the target embedding Δ ahead. Zero means gating it away costs nothing. Synthetic, HGLP-Reg, 2,500 steps, one seed, ground-truth factors.

With the receptive-field bound of Eq. 1 the difference reaches zero at $\Delta = \tau$ and stays there. With the cumulative target of the earlier design — the causal encoder read at $t + \Delta$ — it never reaches zero at any horizon, because that target's receptive field runs back to the start of the window and therefore re-contains the anchor's own transient state. The bounded target also leaves less of the transient factor in z_{per} (R^2 0.041 against 0.660) and does not partially collapse (target RankMe 12.9 against 5.3). Two limits: this measures what explains the target embedding rather than downstream error, and it reports only the *increment* the transient factor adds, so it does not by itself establish that the persistent factor explains the target well.

The target is latent, not raw. Holding gate, anchor, horizon and loss form fixed and predicting the raw patch instead of the encoder's reading of it halves SEP on synthetic data: 0.809 against 0.415 on HGLP-Reg and 0.844 against 0.432 on HGLP-NCE, each at its own best λ, $n = 5$ seeds. Allocation is the same for the two targets, at 0.86 to 0.89, and the gap opens in inclusion — the raw target loses the persistent factor as the penalty grows, from 0.807 to 0.401, whereas the latent target stays above 0.97. The raw signal still contains the transient component, and the pressure to predict that too pushes the persistent factor out.

B Term decomposition and the non-linear probe

Table 6: Where the shortfall against SFA sits. Cells are *ours minus SFA* on one term of SEP, paired by seed then averaged, $n = 5$; positive means we lead, and our column is taken at the listed λ.

Dataset	Stem	λ	Inclusion	Allocation	Exclusion	SEP
Synthetic	HGLP-Reg	1	$+0.102$	$+0.236$	$+0.156$	$+0.395$
Synthetic	HGLP-NCE	16	-0.006	$+0.327$	$+0.266$	$+0.451$
PTB-XL	HGLP-Reg	4	-0.007	$+0.048$	$\mathbf{-0.136}$	-0.073
PTB-XL	HGLP-NCE	4	-0.000	$+0.007$	$\mathbf{-0.051}$	-0.031
HAPT	HGLP-Reg	1	-0.033	-0.011	-0.007	-0.039
HAPT	HGLP-NCE	64	-0.041	$+0.155$	-0.016	$+0.100$

Table 6 splits SEP into its three factors and pairs by seed. Inclusion is a draw everywhere; every real-data loss against SFA is a loss on exclusion alone, and every win is a win on allocation.

Table 7: What survives a non-linear reader. The features, the splits and the blocks are held fixed and only the probe head changes, from a linear model to a one-hidden-layer MLP (256 units); the *same* head is given to SFA, PCA and ICA, so no method is tested more harshly than another. $\lambda = 4$, one seed. **(a)** Cells are MLP − linear on one SEP term, averaged over four methods × six settings. Only exclusion falls, and it is the term the objective never enforced. **(b)** Cells are SFA's exclusion minus ours, so positive means SFA holds the lead; the diagnosis of Table 6 was not an artifact of linear probing.

(a) Change in each SEP term when the probe head stops being linear

SEP term	Mean change	Worst	Best	Rows
Inclusion	$+0.015$	-0.027	$+0.113$	16
Allocation	$+0.087$	$+0.031$	$+0.156$	16
Exclusion	-0.149	-0.530	$+0.019$	16

(b) Exclusion gap, SFA minus ours

Setting	Linear probe	MLP probe
PTB-XL/HGLP-Reg	$+0.085$	$+0.122$
PTB-XL/HGLP-NCE	$+0.020$	$+0.033$
HAPT/HGLP-Reg	$+0.009$	$+0.236$
HAPT/HGLP-NCE	$+0.000$	$+0.000$

Table 7 replaces the linear reading head with a one-hidden-layer MLP, giving the same head to every method. Part (a) reports the change in each SEP term: exactly one collapses, and it is exclusion. Part (b) reports SFA's exclusion lead under both heads: it survives and in two settings widens, so the diagnosis is not an artifact of linear probing. PCA and ICA degrade sharply under the same head while SFA does not.

Three cautions. Because exclusion falls monotonically as the reading head grows, only comparisons made under a fixed head are meaningful. We tried one hidden width, 256, and did not trace the curve. And this experiment fixes $\lambda = 4$ with one seed, whereas Table 6 takes each setting at its own best λ, so cells of the two tables must not be read against each other.

C ADDITIONAL FIGURES

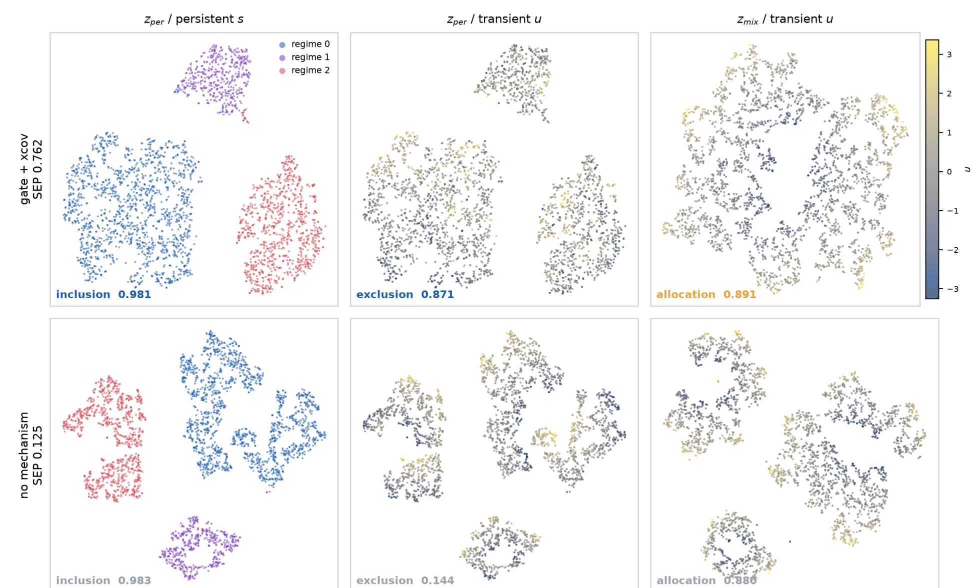

Figure 6: The two blocks against both factors, with and without the mechanism (synthetic $\pm 3\%$, HGLP-NCE, $\lambda = 4$, one seed). Fig. 3 shows the middle column of this figure, which is the only one that differs between the rows. The fourth combination, z_{mix} against the persistent factor, is not a term of SEP and is unconstrained by design, so it is omitted here too.

The persistent block can also be read as a signal in its own right rather than as input to a classifier. Fig. 7 plots its leading principal components against time on a held-out HAPT subject, with the activity boundaries marked.

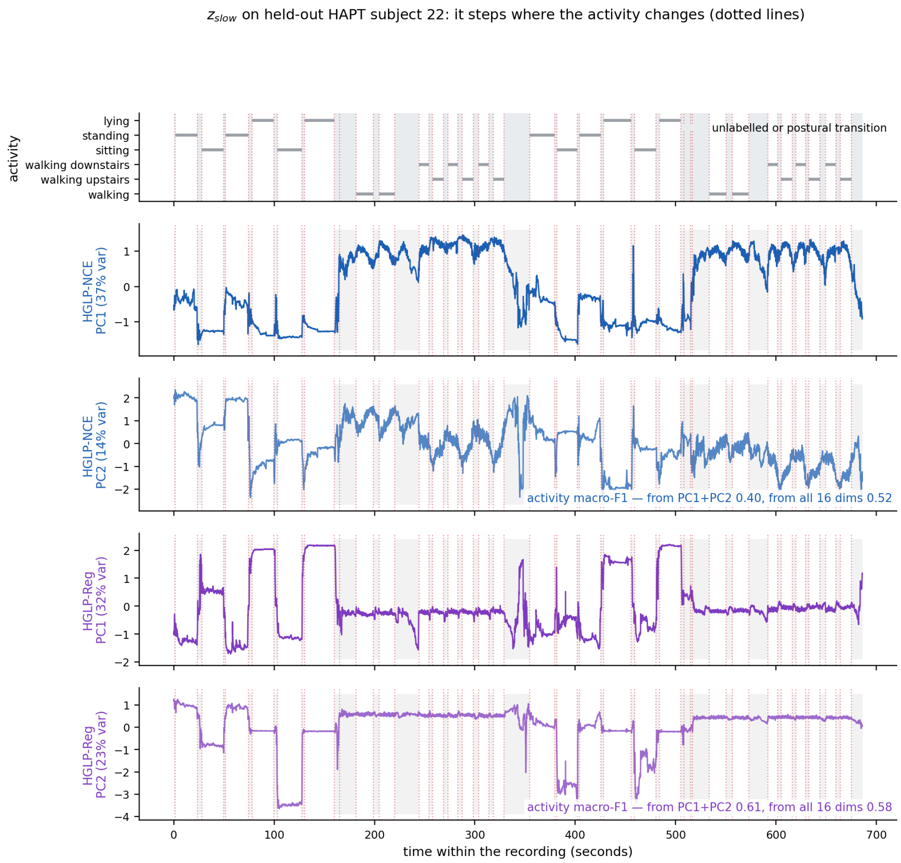

Figure 7: z_{per} read as a trajectory rather than as classifier input (HAPT). This use needs no labels at evaluation time and no probe. The figure is qualitative.